

Regime & Risk: Energy Benchmarks

WTI, Brent, Henry Hub and an \$8mm energy book · daily data 2007–2026-09-22 · source: GitHub mirror of EIA · generated 2026-09-27

Headline numbers

BOOK 1-DAY 99% VAR

\$258.4k

GARCH-t, as of 2026-09-22

BOOK 97.5% ES

\$274.1k

average loss beyond VaR

BEST BACKTEST RECORD

**Filtered HS &
GARCH-t**

each passes 3 of 4 series

HS MISSES IN
TURBULENCE

3.0%

vs 0.5% in calm; target 1.0%

BEST VARIANCE
FORECAST

GARCH-t

lowest QLIKE on 4 of 4 series

TURBULENT NOW

none

regime with highest filtered
probability

- **Regimes are real and persistent.** Calm spells last 51–94 trading days on average; turbulent spells 4–13 days, with volatility 3.5–6.9× the calm level.
- **Plain historical simulation is regime-blind.** Its 99% VaR was beaten on 0.5% of calm days but 3.0% of turbulent days: the 250-day window is too slow to absorb a new volatility regime, so exceptions arrive in clusters (Christoffersen rejects HS on 3 of 4 series).
- **Filtering fixes most of it.** Rescaling history to today's EWMA volatility (filtered HS) cuts the turbulent-regime exception rate to 0.6% and passes coverage and independence on 3 of 4 series.
- **GARCH-t forecasts variance best, yet its VaR still fails on WTI:** 62 exceptions against 43 expected (Kupiec $p = 0.005$). A good volatility forecast is necessary for a good VaR, not sufficient: the tail shape matters too.
- **Normal tails fail.** HAR-RV with normal quantiles records 268 exceptions across the four series against 171 expected; its average Z2 of -0.29 says realised tail losses run well beyond its ES.

Note. All forecasts are out of sample: each day's VaR uses only data available the evening before. The positions are hypothetical and priced off EIA spot benchmarks, not tradable futures.

1. Data

Daily EIA spot prices from 2007-01-02 to 2026-09-22. Returns are daily log changes in %. The **energy book** holds constant dollar positions: WTI \$4.00mm, Brent -\$2.00mm, Henry Hub \$2.00mm (a long WTI–Brent spread plus long US gas); its return is P&L over gross notional.

Data pull log

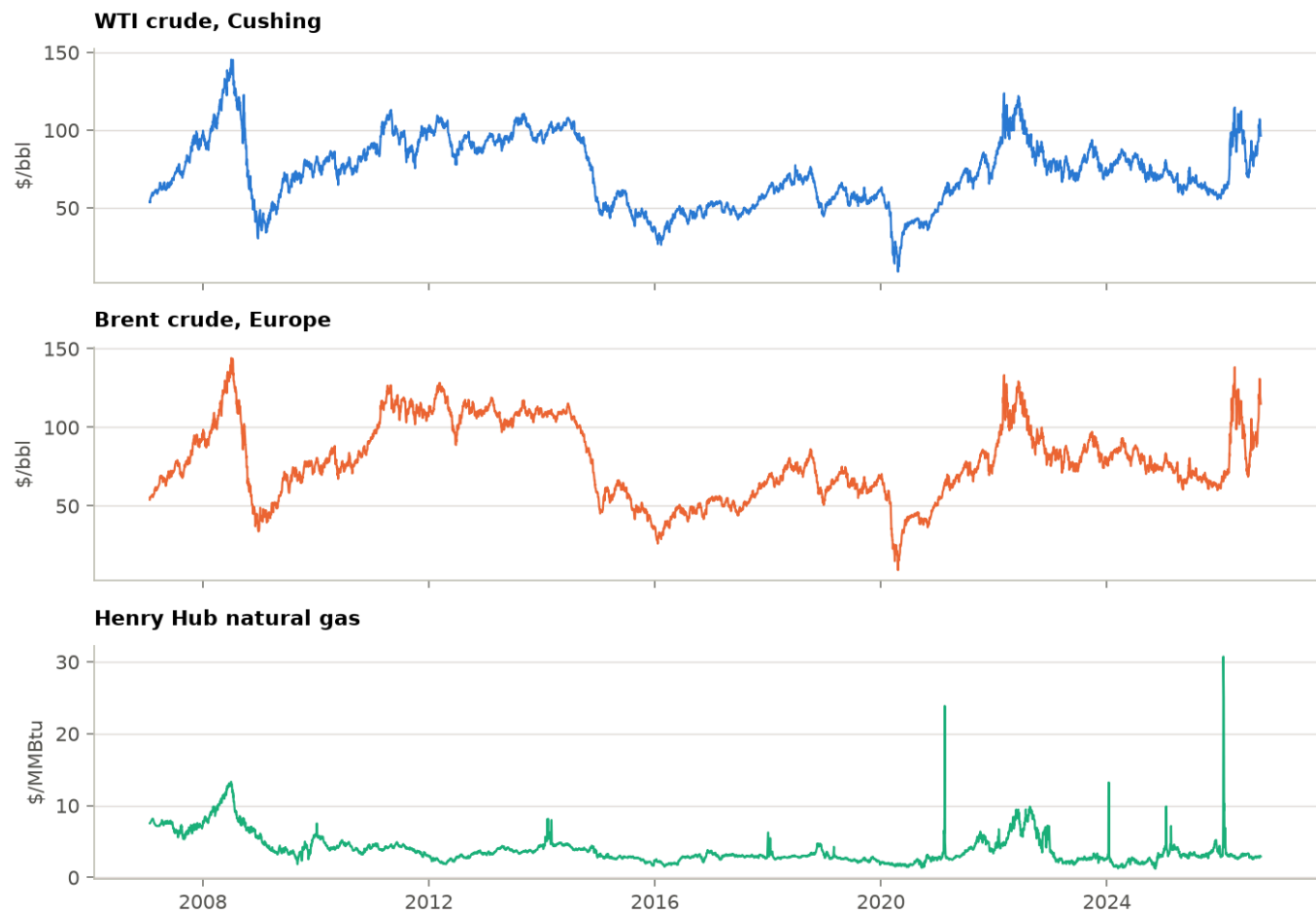
Series	Description	Unit	Source	First	Last	Rows
WTI	WTI crude, Cushing	\$/bbl	GitHub mirror of EIA (DCOILWTICO)	1986-01-02	2026-09-22	9,507
Brent	Brent crude, Europe	\$/bbl	GitHub mirror of EIA (DCOILBRETEU)	1987-05-20	2026-09-22	9,092
HenryHub	Henry Hub natural gas	\$/MMBtu	GitHub mirror of EIA (DHHNGSP)	1997-01-07	2026-09-22	5,692

Return statistics (daily log returns)

Series	Ann. vol	Skew	Excess kurtosis	Worst day	Best day
WTI	47.6%	-2.17	89.2	-72.0% (2020-04-21)	42.6% (2020-04-22)
Brent	43.8%	-2.07	80.0	-64.4% (2020-04-21)	41.2% (2020-04-22)
HenryHub	111.5%	1.17	117.3	-140.2% (2024-01-16)	143.3% (2024-01-12)
Book	39.5%	11.95	338.2	-19.2% (2024-01-16)	80.4% (2024-01-12)

Data quality. 5 fixes were applied and logged (`01_data_quality_fixes.csv`), including WTI's negative print on 2020-04-20 (set to missing: a log return is undefined). 45 extreme moves were **flagged but kept** (`01_flagged_moves.csv`); crashes and gas spikes are exactly what risk models must survive.

Daily spot prices (EIA)



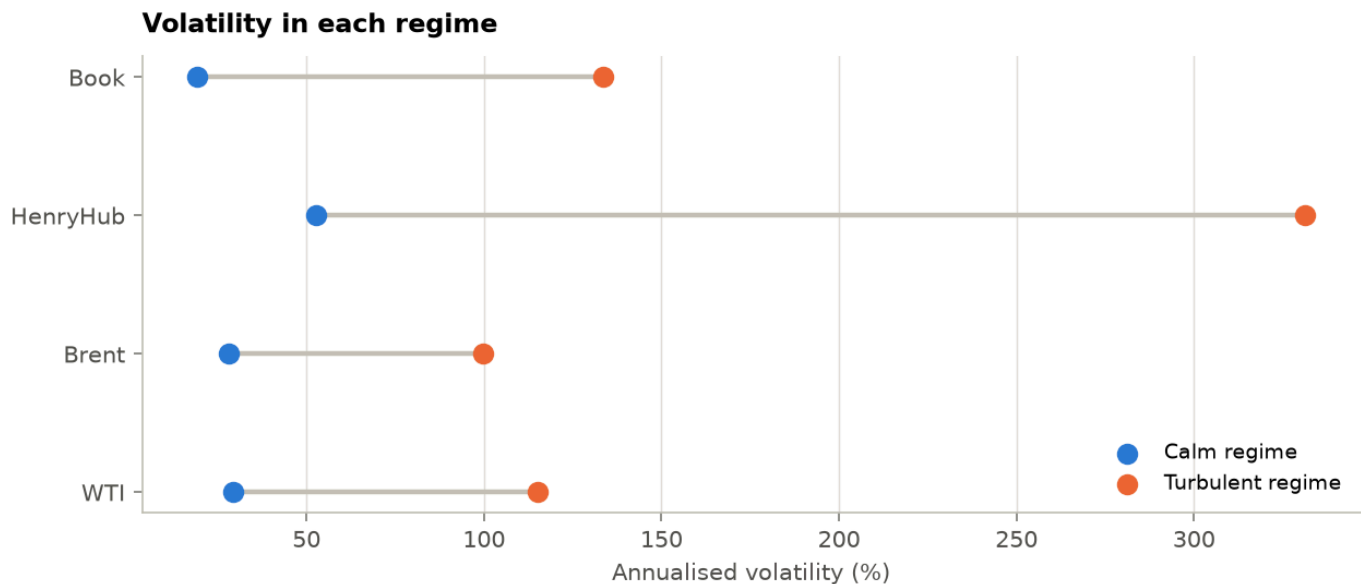
EIA daily spot prices.

2. Regimes

A Gaussian **Markov-switching model** (Hamilton, 1989) lets each market move between a calm and a turbulent state, each with its own mean and volatility, switching with fixed daily probabilities. The regime is never observed; the model infers the probability of each regime every day.

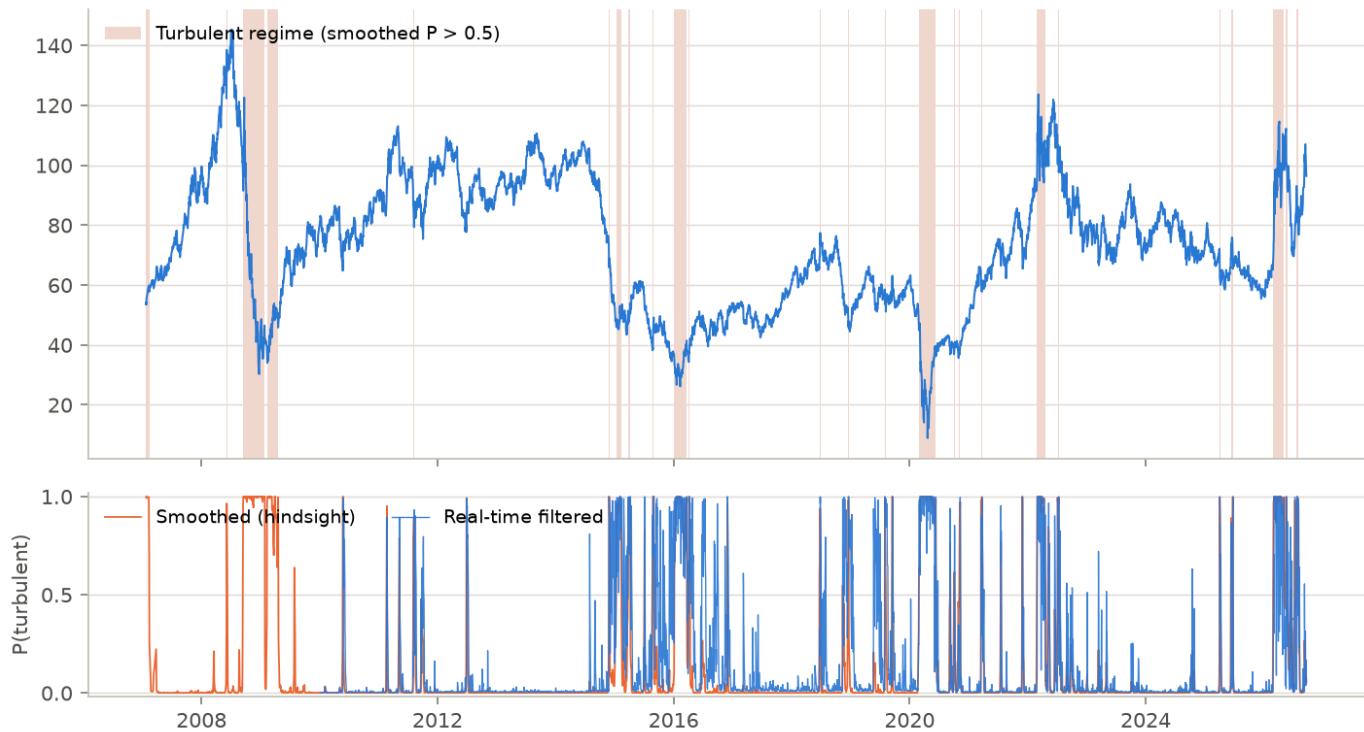
Two-regime model, full sample

Series	Regime	Ann. vol	Daily mean	Stay prob.	Avg. spell (days)	Share of days
WTI	Calm	29.4%	0.032%	0.989	91.2	89.8%
WTI	Turbulent	115.2%	-0.147%	0.912	11.4	10.2%
Brent	Calm	28.2%	0.034%	0.989	93.8	88.6%
Brent	Turbulent	99.8%	-0.119%	0.924	13.1	11.4%
HenryHub	Calm	52.6%	-0.039%	0.980	50.8	92.0%
HenryHub	Turbulent	331.2%	0.179%	0.802	5.0	8.0%
Book	Calm	19.3%	0.006%	0.982	55.4	94.3%
Book	Turbulent	133.5%	1.065%	0.748	4.0	5.7%



Annualised volatility in each regime.

WTI: price with turbulent regimes shaded



WTI: turbulent regimes (shaded) and the probability of turbulence, smoothed vs real time.

Model choice. BIC prefers three regimes on 4 of 4 series (e.g. WTI: 22,341 with two vs 21,930 with three). With Gaussian regimes an extra state mostly absorbs fat tails, so this project keeps the two-regime model for interpretability and reports the three-regime fit in `02_model_selection.csv`. That trade-off is a documented model choice, not an oversight.

Does the estimator work? Three data sets were simulated from each fitted model and re-estimated. The EM algorithm recovers regime volatilities within a few percent and classifies the true hidden state on almost every day:

Simulation-based verification

Series	Worst vol error	Worst state accuracy
Book	4.9%	98.5%
Brent	5.8%	98.0%
HenryHub	3.5%	97.8%
WTI	6.2%	98.1%

Current regime (filtered, 2026-09-22)

Series	Regime today	Probability	$P(\text{turbulent})$	Days in regime
WTI	Calm	0.94	0.06	35
Brent	Calm	0.56	0.44	1
HenryHub	Calm	1.00	0.00	52
Book	Calm	1.00	0.00	117

Longest turbulent episodes (smoothed probability > 0.5)

Series	Start	End	Days	Price change	Realised vol
Brent	2008-09-15	2009-01-27	93	-52.6%	82%
WTI	2008-09-15	2009-01-27	93	-56.4%	99%
Brent	2026-03-02	2026-06-29	84	-7.3%	81%
WTI	2020-02-28	2020-06-11	73	-18.7%	239%
Brent	2020-03-06	2020-06-12	69	-15.5%	213%
WTI	2016-01-06	2016-03-23	56	12.7%	74%
Brent	2022-02-25	2022-05-12	54	9.6%	71%
WTI	2009-02-10	2009-04-20	48	22.1%	82%

3. Volatility forecasts

Four one-day-ahead variance forecasts, all out of sample. They are scored with **QLIKE** ($\log h + r^2/h$), which ranks forecasts correctly even though the squared return is a very noisy measure of true variance (Patton, 2011). **Diebold–Mariano** tests ask whether the gap to GARCH-t is bigger than chance.

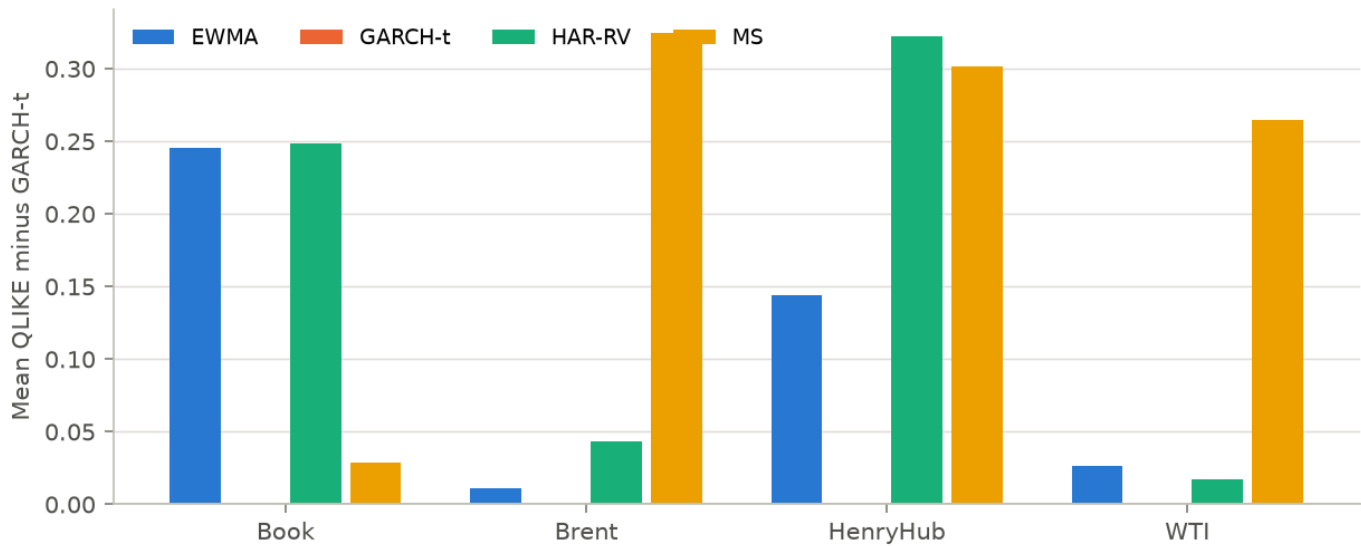
Mean QLIKE loss (lower is better; ★ = best)

Series	EWMA	GARCH-t	HAR-RV	Markov-switching
Book	2.446	2.201 ★	2.449	2.229
Brent	2.425	2.414 ★	2.457	2.739
HenryHub	4.149	4.004 ★	4.327	4.306
WTI	2.541	2.514 ★	2.531	2.778

Diebold–Mariano tests against GARCH-t (positive = higher loss than GARCH-t)

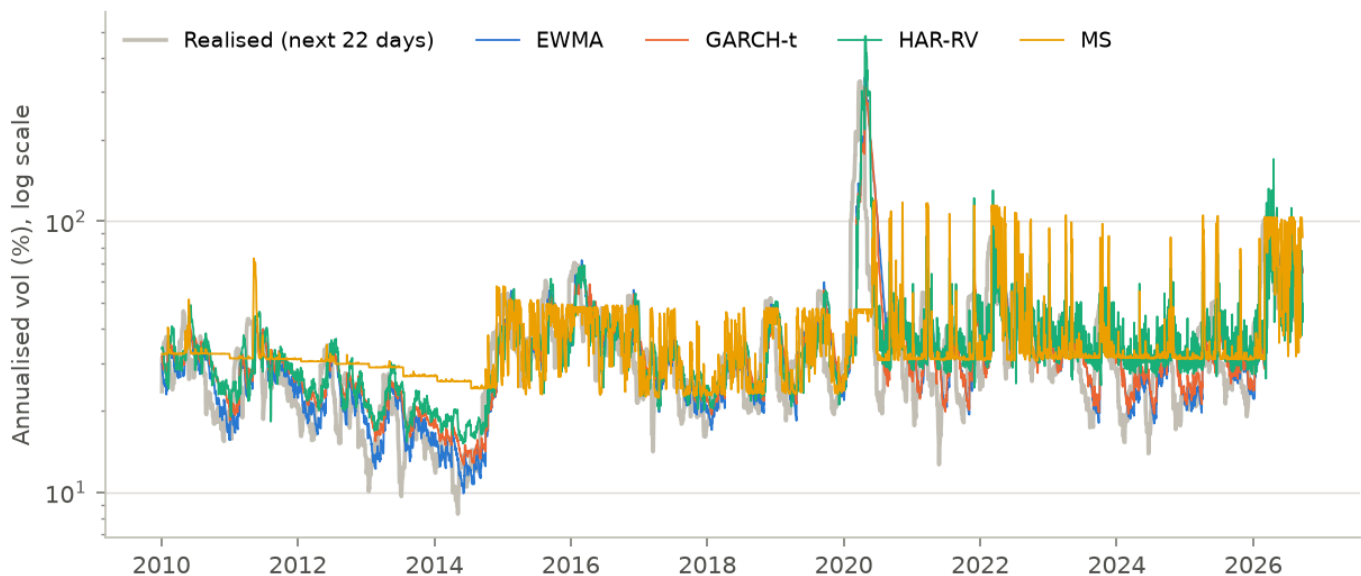
Series	Model vs GARCH-t	DM stat (QLIKE)	p-value	Verdict
WTI	EWMA	3.71	<0.001	✖ Worse than GARCH-t
WTI	HAR-RV	1.60	0.110	No significant difference
WTI	MS	2.17	0.030	✖ Worse than GARCH-t
Brent	EWMA	1.89	0.058	No significant difference
Brent	HAR-RV	3.14	0.002	✖ Worse than GARCH-t
Brent	MS	2.00	0.046	✖ Worse than GARCH-t
HenryHub	EWMA	2.84	0.005	✖ Worse than GARCH-t
HenryHub	HAR-RV	2.32	0.020	✖ Worse than GARCH-t
HenryHub	MS	3.03	0.002	✖ Worse than GARCH-t
Book	EWMA	1.90	0.057	No significant difference
Book	HAR-RV	1.59	0.111	No significant difference
Book	MS	0.26	0.793	No significant difference

Forecast loss relative to GARCH-t (below zero = better)



Loss relative to GARCH-t by series.

Brent: one-day-ahead volatility forecasts



Brent: forecasts vs the volatility realised over the next month.

GARCH estimator check. Refitting simulated GARCH-t paths recovers persistence ($\alpha + \beta$) within 0.015 on every series. HAR-RV is handicapped here: with daily closes only, realised variance is proxied by the squared return, which throws away most of the information intraday data would give.

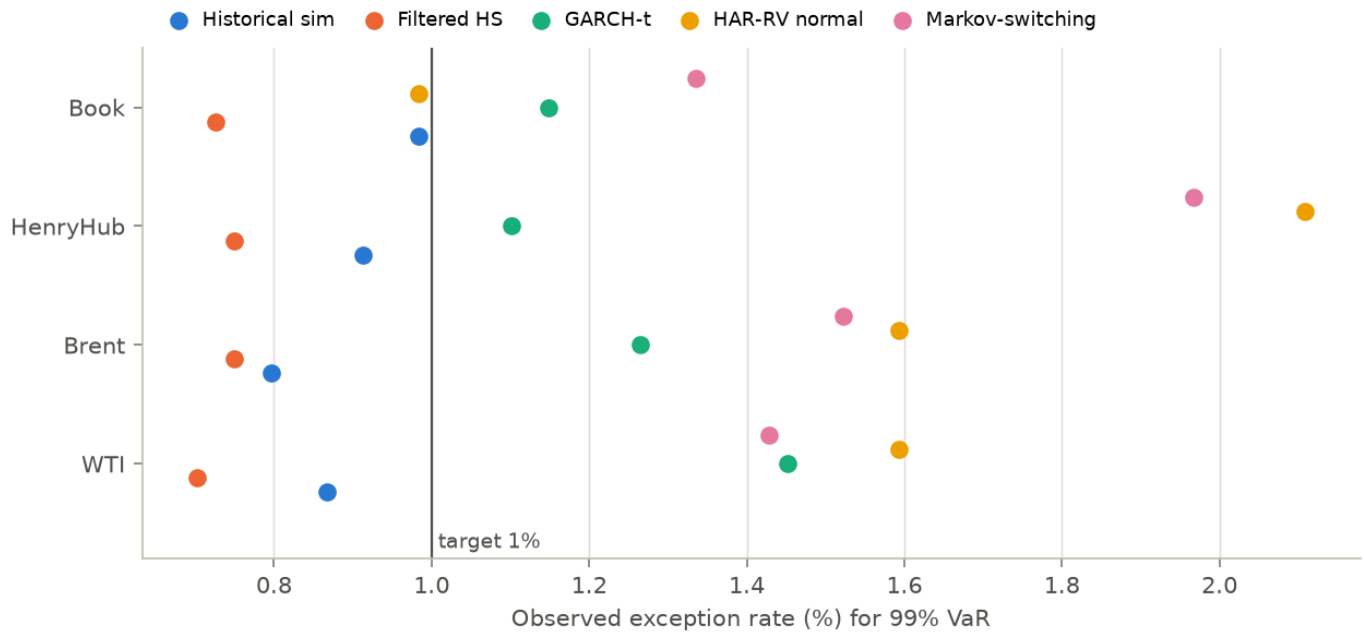
4. VaR and ES backtests

Five models forecast 1-day 99% VaR and 97.5% ES every day from 2010-01-04 (4,269 days). A model **passes** when neither Kupiec (right number of exceptions) nor conditional coverage (right number *and* no clustering) rejects at 5%.

Backtest results, 99% VaR (exceptions observed / expected)

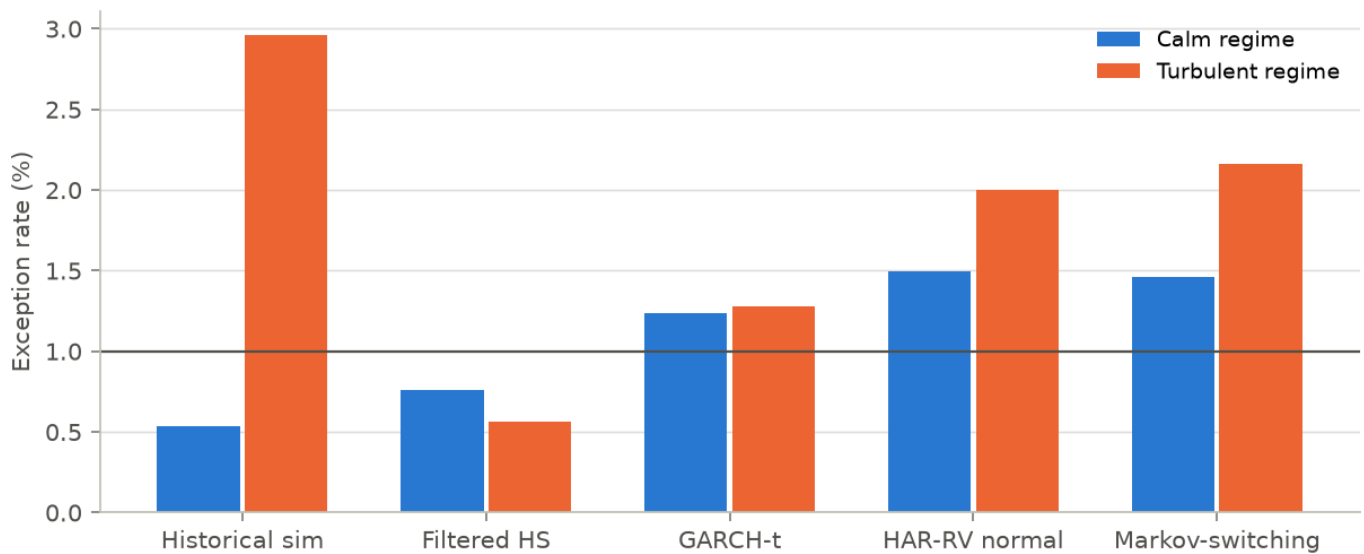
Series	Model	Exceptions	Kupiec p	Christoffersen p	CC p	Zone (last 250d)	Z2	Verdict
WTI	Historical sim	37 / 43	0.370	0.004	0.010	green	−0.19	✗ Reject
WTI	Filtered HS	30 / 43	0.039	0.018	0.007	green	−0.06	✗ Reject
WTI	GARCH-t	62 / 43	0.005	0.310	0.012	green	−0.38	✗ Reject
WTI	HAR-RV normal	68 / 43	<0.001	0.934	0.002	yellow	−0.27	✗ Reject
WTI	Markov-switching	61 / 43	0.008	0.892	0.030	green	−0.41	✗ Reject
Brent	Historical sim	34 / 43	0.166	0.276	0.212	green	−0.24	✓ Pass
Brent	Filtered HS	32 / 43	0.085	0.242	0.115	green	−0.09	✓ Pass
Brent	GARCH-t	54 / 43	0.095	0.716	0.232	green	−0.20	✓ Pass
Brent	HAR-RV normal	68 / 43	<0.001	0.422	0.001	yellow	−0.33	✗ Reject
Brent	Markov-switching	65 / 43	0.001	0.097	0.002	green	−0.33	✗ Reject
HenryHub	Historical sim	39 / 43	0.565	<0.001	<0.001	green	−0.31	✗ Reject
HenryHub	Filtered HS	32 / 43	0.085	0.487	0.179	green	0.03	✓ Pass
HenryHub	GARCH-t	47 / 43	0.514	0.306	0.479	green	−0.06	✓ Pass
HenryHub	HAR-RV normal	90 / 43	<0.001	0.172	<0.001	green	−0.67	✗ Reject
HenryHub	Markov-switching	84 / 43	<0.001	0.113	<0.001	yellow	−0.59	✗ Reject
Book	Historical sim	42 / 43	0.915	<0.001	<0.001	yellow	−0.11	✗ Reject
Book	Filtered HS	31 / 43	0.059	0.501	0.133	green	−0.00	✓ Pass
Book	GARCH-t	49 / 43	0.343	0.594	0.554	green	0.04	✓ Pass
Book	HAR-RV normal	42 / 43	0.915	0.435	0.733	green	0.11	✓ Pass
Book	Markov-switching	57 / 43	0.036	0.230	0.054	yellow	−0.24	✗ Reject

How often did losses beat VaR?



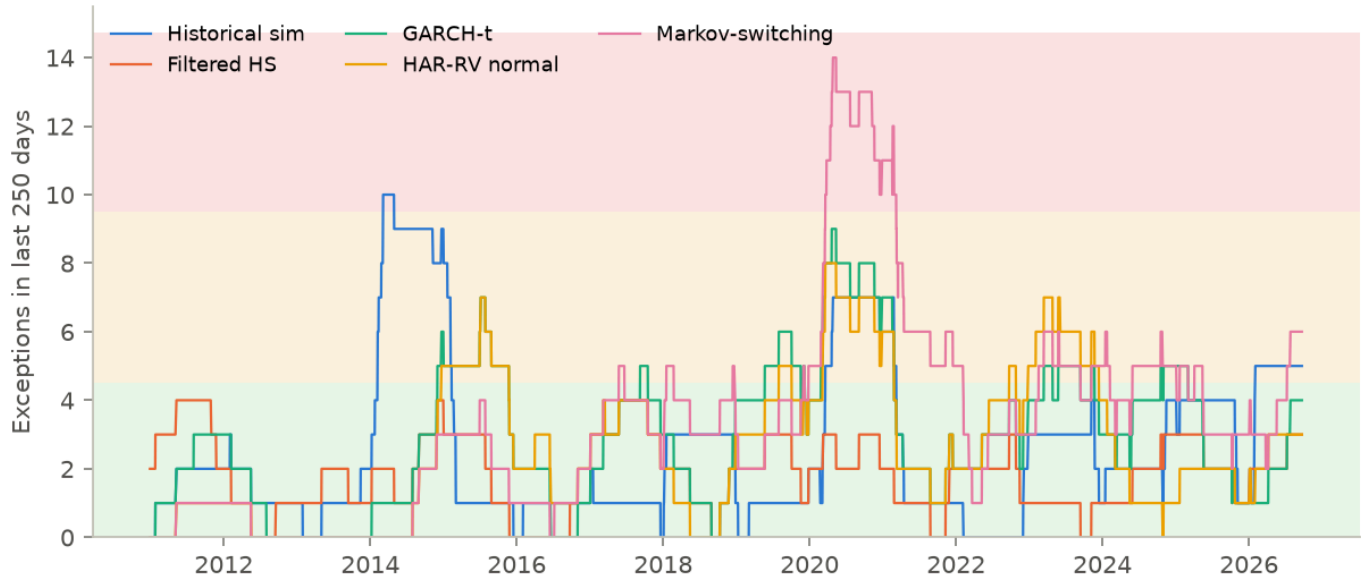
Observed exception rates against the 1% target.

Exception rate by real-time regime (all series pooled; target 1%)



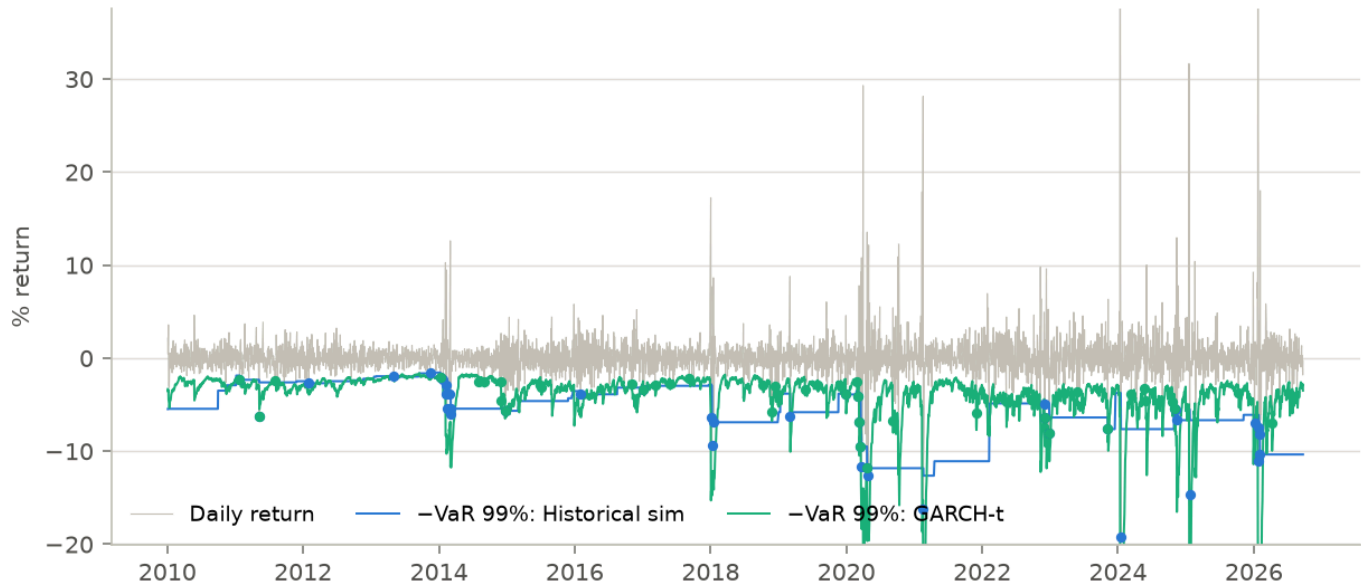
Exception rate split by the regime the model knew about at forecast time.

Energy book: Basel traffic light over time (green 0-4, yellow 5-9, red 10+)



Rolling 250-day exception counts for the energy book with Basel zones.

Book: daily returns vs VaR forecasts (dots = exceptions)



Energy book: returns against historical-simulation and GARCH-t VaR.

Next-day risk as of 2026-09-22

Series	Model	Notional	VaR 99%	ES 97.5%	VaR % of notional
WTI	Historical sim	\$1.00mm	\$115.6k	\$109.6k	12.28%
WTI	Filtered HS	\$1.00mm	\$82.3k	\$79.8k	8.58%
WTI	GARCH-t	\$1.00mm	\$69.3k	\$71.7k	7.18%
WTI	HAR-RV normal	\$1.00mm	\$55.6k	\$55.9k	5.72%
WTI	Markov-switching	\$1.00mm	\$70.6k	\$74.0k	7.32%

Series	Model	Notional	VaR 99%	ES 97.5%	VaR % of notional
Brent	Historical sim	\$1.00mm	\$123.5k	\$109.8k	13.19%
Brent	Filtered HS	\$1.00mm	\$118.0k	\$101.6k	12.55%
Brent	GARCH-t	\$1.00mm	\$94.5k	\$97.8k	9.93%
Brent	HAR-RV normal	\$1.00mm	\$66.4k	\$66.7k	6.87%
Brent	Markov-switching	\$1.00mm	\$117.7k	\$118.2k	12.53%
HenryHub	Historical sim	\$1.00mm	\$387.2k	\$338.2k	48.97%
HenryHub	Filtered HS	\$1.00mm	\$117.4k	\$82.8k	12.49%
HenryHub	GARCH-t	\$1.00mm	\$80.9k	\$87.2k	8.43%
HenryHub	HAR-RV normal	\$1.00mm	\$122.1k	\$122.6k	13.02%
HenryHub	Markov-switching	\$1.00mm	\$88.6k	\$118.2k	9.27%
Book	Historical sim	\$8.00mm	\$828.8k	\$680.9k	10.36%
Book	Filtered HS	\$8.00mm	\$294.4k	\$257.0k	3.68%
Book	GARCH-t	\$8.00mm	\$258.4k	\$274.1k	3.23%
Book	HAR-RV normal	\$8.00mm	\$433.4k	\$435.5k	5.42%
Book	Markov-switching	\$8.00mm	\$259.0k	\$350.9k	3.24%

Expected Shortfall. Z2 near zero means ES was about right; negative values mean realised losses beyond VaR were larger than ES predicted. Acerbi and Szekely's thresholds (−0.70 yellow, −1.8 red) are calibrated to one year of data, so the share of rolling years below −0.70 is in 04_backtest_results.csv .

5. Model validation summary

Organised the way a model risk team reviews a model under the Fed/OCC **SR 11-7** guidance.

Validation evidence

Check	How	Result	Status
Conceptual soundness	Simulate from each fitted model, re-estimate, compare	MS vol error $\leq 6.2\%$, state accuracy $\geq 97.8\%$; GARCH persistence error ≤ 0.015	✓ Pass
Model selection	BIC across 2 vs 3 regimes	BIC prefers 3 on 4 of 4 series; 2 kept for interpretability	▲ Documented choice
Benchmarking	Diebold–Mariano against GARCH-t	GARCH-t has the lowest QLIKE on 4 of 4 series	✓ Pass
Outcomes analysis	Kupiec, Christoffersen, conditional coverage, Z2, McNeil–Frey	8 of 20 model–series pairs pass; best: Filtered HS & GARCH-t	▲ Mixed
Ongoing monitoring	Basel traffic light on rolling 250 days	15 of 20 pairs green today	✓ Pass

Limitations

- Spot benchmarks, not futures: Henry Hub spot spikes during cold snaps are far larger than front-month futures moves.
- Daily closes only: HAR-RV and realised-volatility checks use squared returns, a noisy proxy.
- Gaussian regimes: tails inside each regime are thin; a Student-t Markov-switching model would be the next step.
- Univariate book: the book's risk is modelled on its own P&L series, so correlations change only implicitly.
- Positions are constant-dollar and linear: no options, no liquidity horizons, no basis between spot and hedges.

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